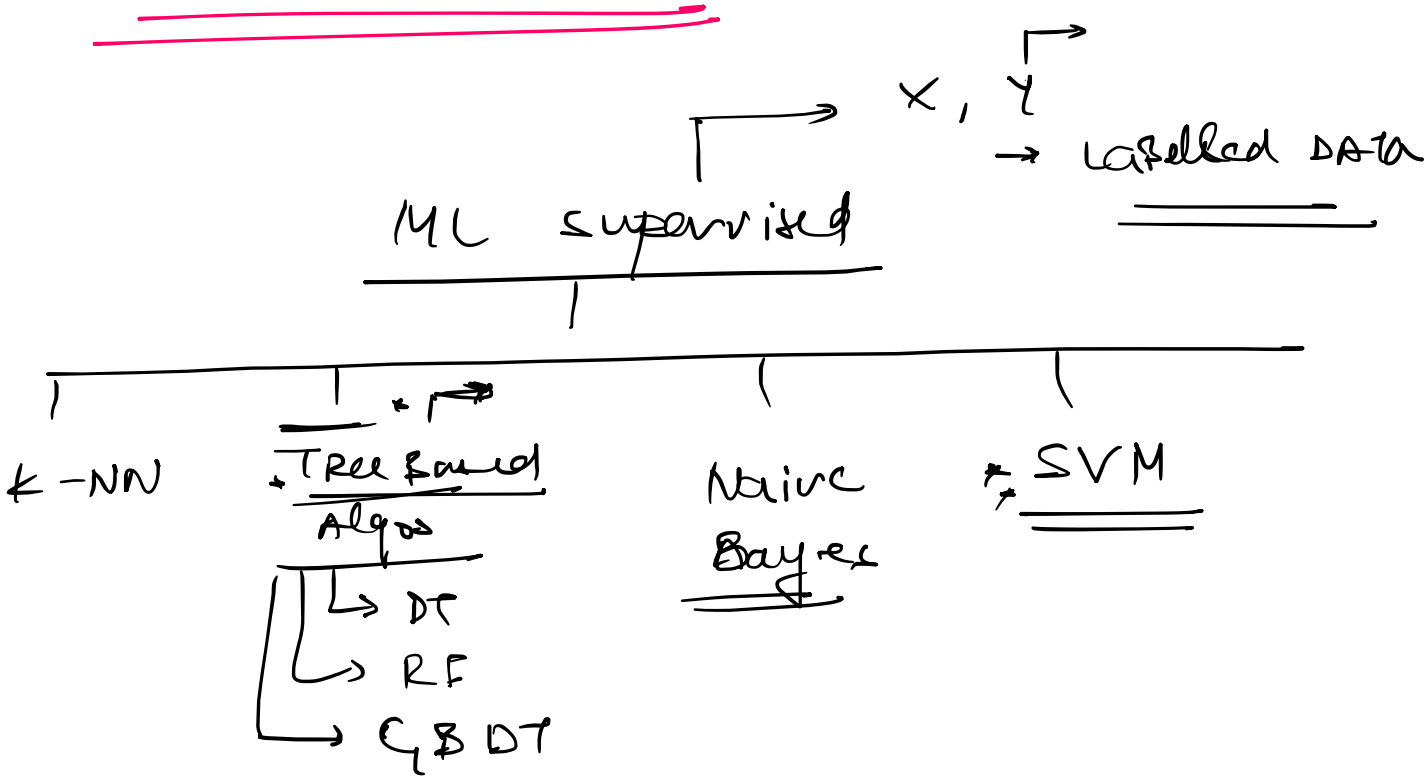
* K NEAREST NEIGHBOR

* Agenda ⇒

(*) Blinkit ⇒

→ * KNN ⇒

→ * CROSS Validation (*)



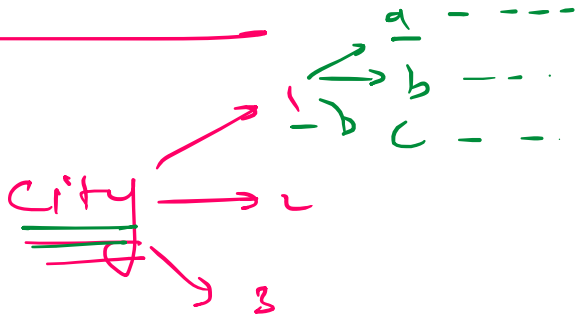
KNN → (*) simple algo ⇒

(*) Blinkit

(*) Blinkit

Problem statement ⇒

(*) Traffic ⇒



more traffic

ordering, purchases

(*) Specific areas

TT

Manesar

gurgaon

(*) KNN ⇒

(*) Delivery partners?

→ (*) Classes → 1 → High traffic

→ 2 → Moderate traffic

→ 3 → Low traffic

440 Rows & 8 columns

T i/p , 1 target

apple orange grapes



(*) Solve this type of problem ⇒

→ logistic Reg^n with OVR

one V/S Rest

(k) 3 classes ⇒ 3 models

n classes → n models

I features, Y ⇒ Algo to visualize High dim^n data

X

↓

PCA

n features → 2 or 3 new features

linear comb^n of existing features

linear ...

existing features

→ imb. data ⇒ SMOTE

(*) log Reg ⇒

(1) Non linear → Polynomial features +

(2) Multi class → OVR

10 classes → 10 models

100 classes → 100 models

KNN = (*) [Non linear + Multiclass problems]

SMOTE (*)

imbalance → SMOTE → Train data

Train, test

KNN ⇒

Blinkit store

- cat 1 — High
- cat 2 — Moderate
- cat 3 — low

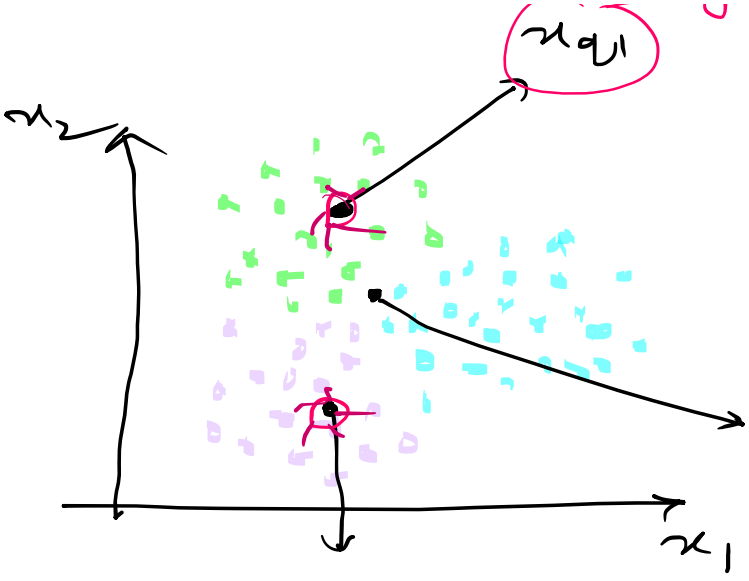
→ Non linear

→ multiclass

→ classification + Regression

Diff. stores info → High, low, moderate





query point ⇒ PP

① x_{q1} → Green

② x_{q2} → purple

x_{q2}
↓
Purple

③ x_{q3} → ??

Concept of Neighborhood ⇒

→ Nearest Neighbors

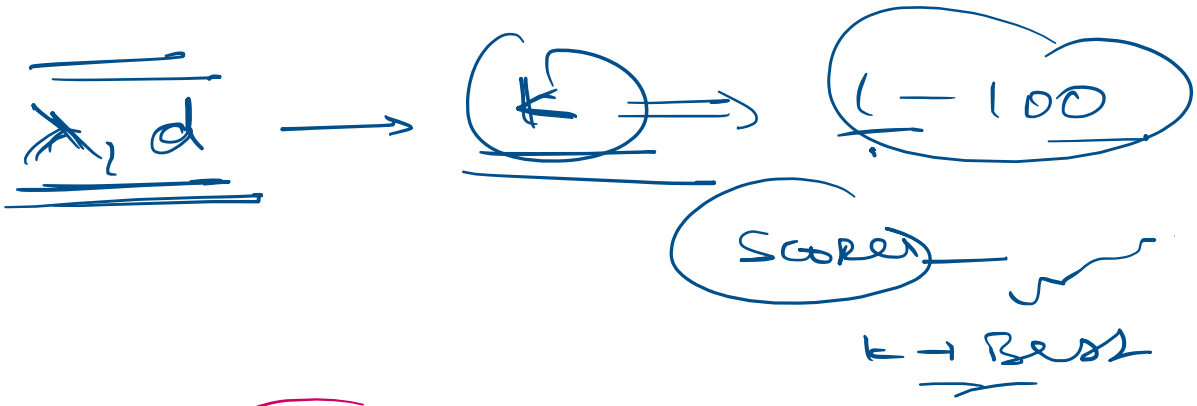
K-NN

→ Hyperparam

we need to decide value of K

④ Hyperparam Tuning

⑤ Class
↓
P1



K = 5 (5) Nearest Neighbor

K = 111 → (11)

20 →

⊛ ⇒

① find Best K ⇒ ② Hyperparam tuning

1 — 100 × 10

Train, val data

↳ find Best K

Quiz time!

Time Left: 35s

How will Logistic Regression handle non-linear,multi-class data ?

33 users have participated

✓	A	Polynomial, OneVsRest	85%
	B	Linear, OneVsRest	9%
	C	OneVsRest, Polynomial	6%
	D	OneVsRest, Linear	0%

End Quiz Now

5 mins

10:02

Email

Medium

Towards ~~DATA SCIENCE~~

OOD

Train Test

Regression → LR

→ KNN

→ DTree

RF

Campus

GenAI

② KNN intuition

③ ALGORITHM

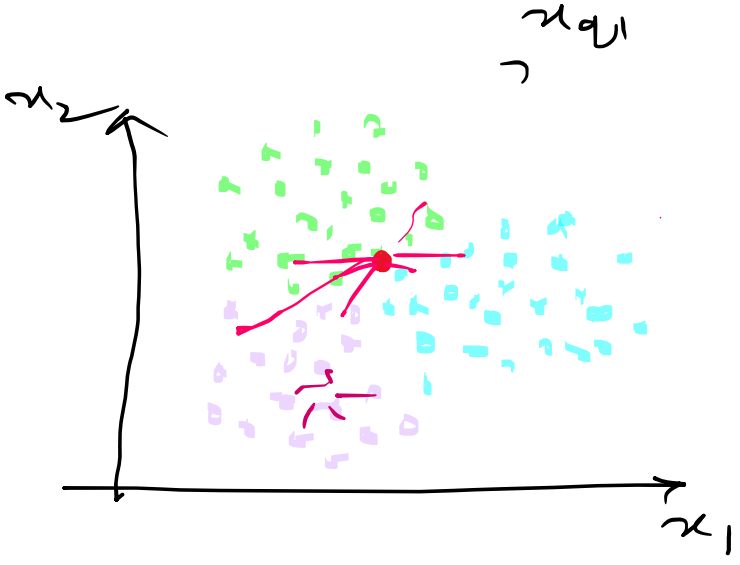
input → (X_train, y_train)

To predict

input → (X_train, y_train) → Train DATA , x query

3 simple steps ⇒

①



● → query PP

① Compute dist. of query point from all of training data

→ dist B/W 2 DP

list of all the distances

(2) SORT distances in ascending orders

[1 2 —]

(3) TOP k = 5 of distances & their label

7, 10

→ (dist, 2nd, 3rd + cat2, 4th, 5th cat1 cat2

(*) cat1 cat1

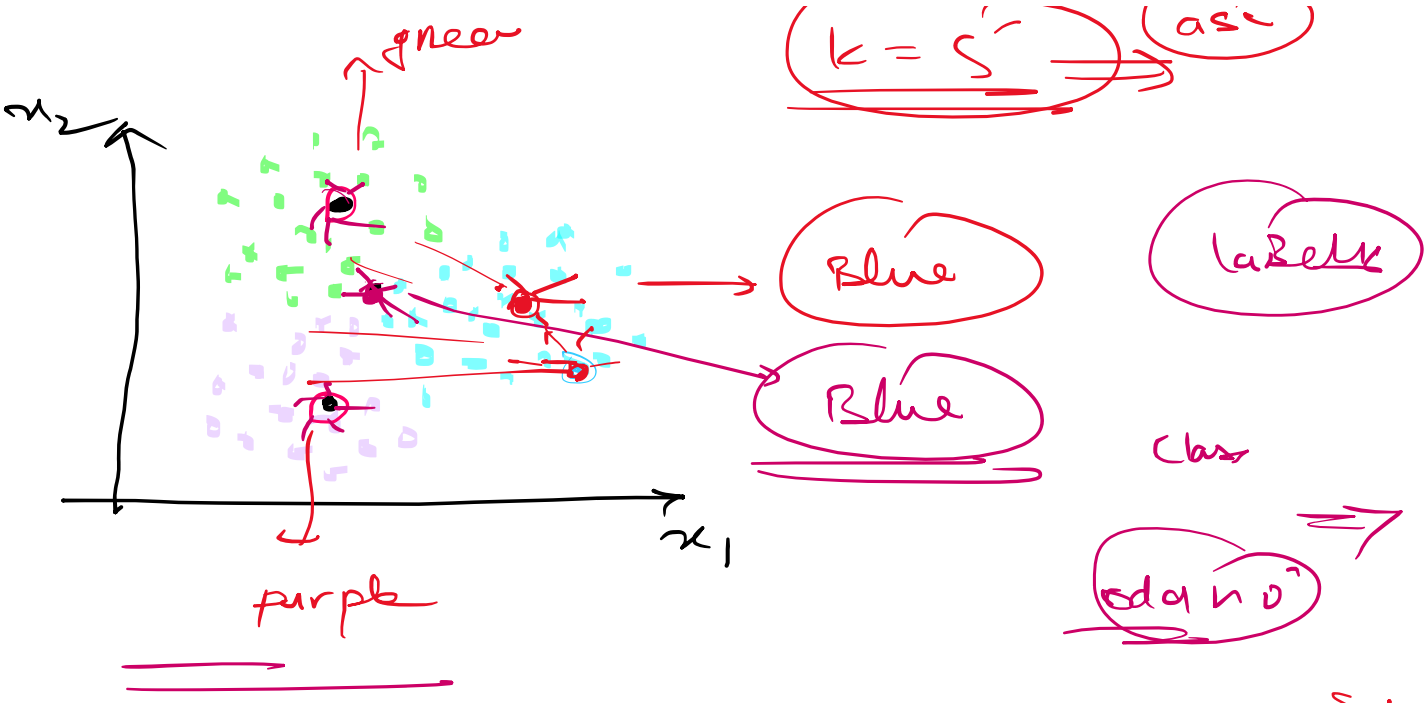
cat1 → (*) → mode

majority vote

(*) label pred^n

idea → ↑ near

k = 5 → ask



Occams Razor → simple

→ No Training involved

KNN → Non parametric

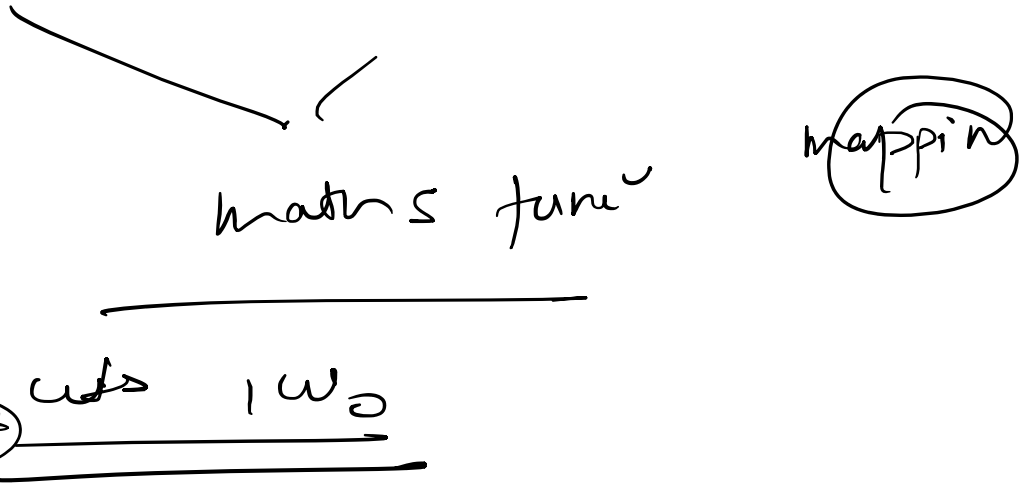
(No params to learn)

(No math func^n to learn)

Parametric algos

Linear

Logistic


maths fun
mappin
cats
KNN
simple logic
distance
sorting
Testing | Inference
processing
happens
KNN does everything
during test |
prediction time
Quiz time!
Time Left: 11s
Arrange the statements in correct order based on kNN algo

Arrange the statements in correct order based on kNN algo

s1- find majority vote

s2- perform euclidean distance

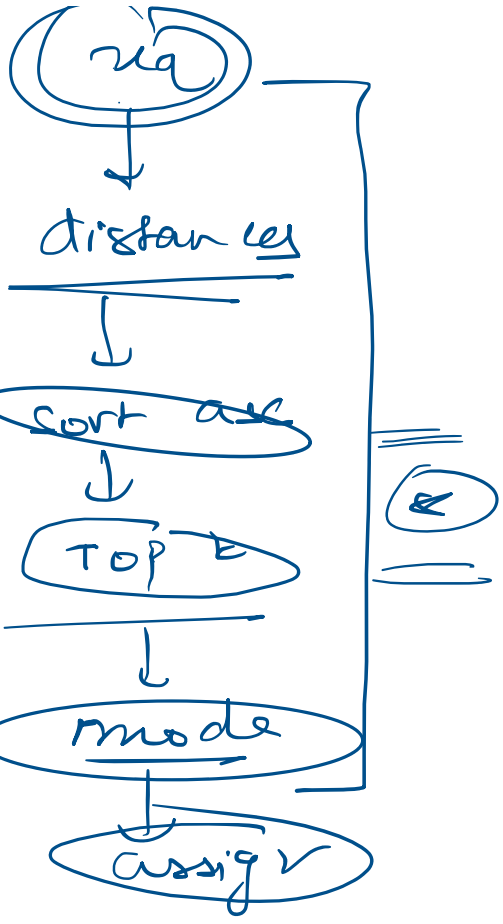
s3- sort and select k datapoints

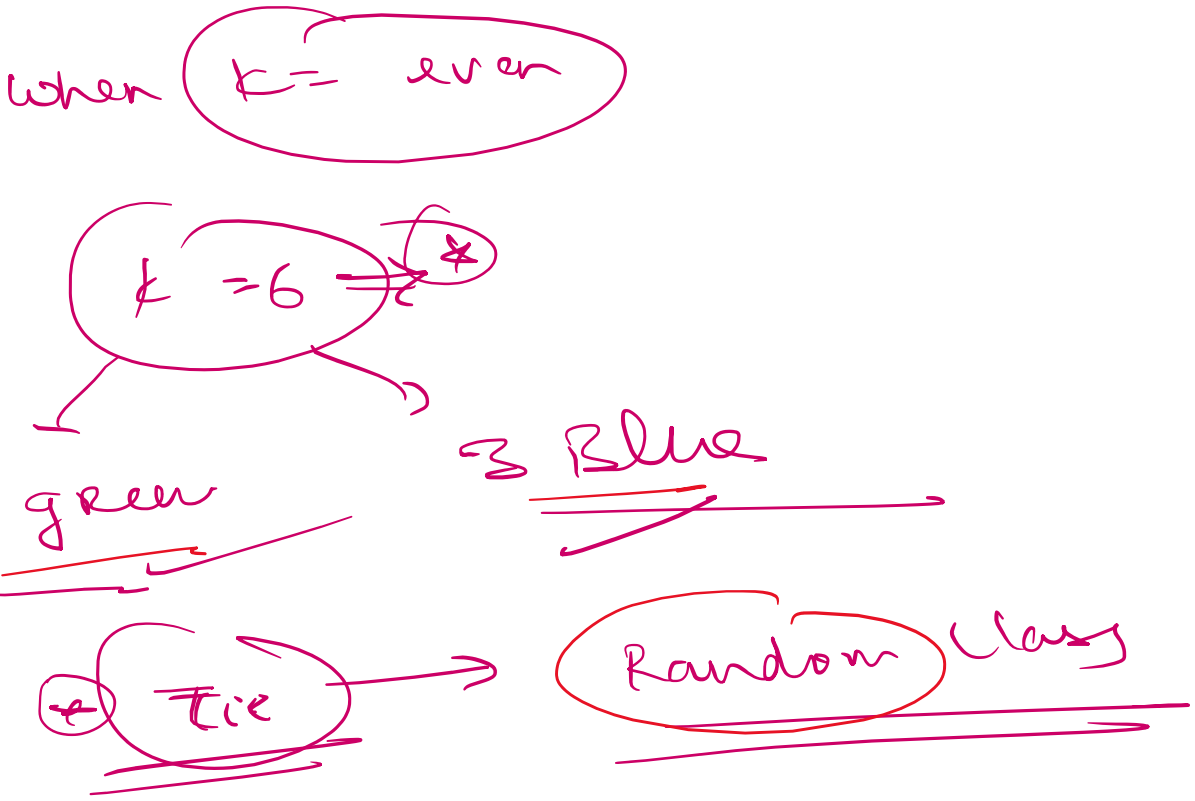
s4- give class to x_q datapoint

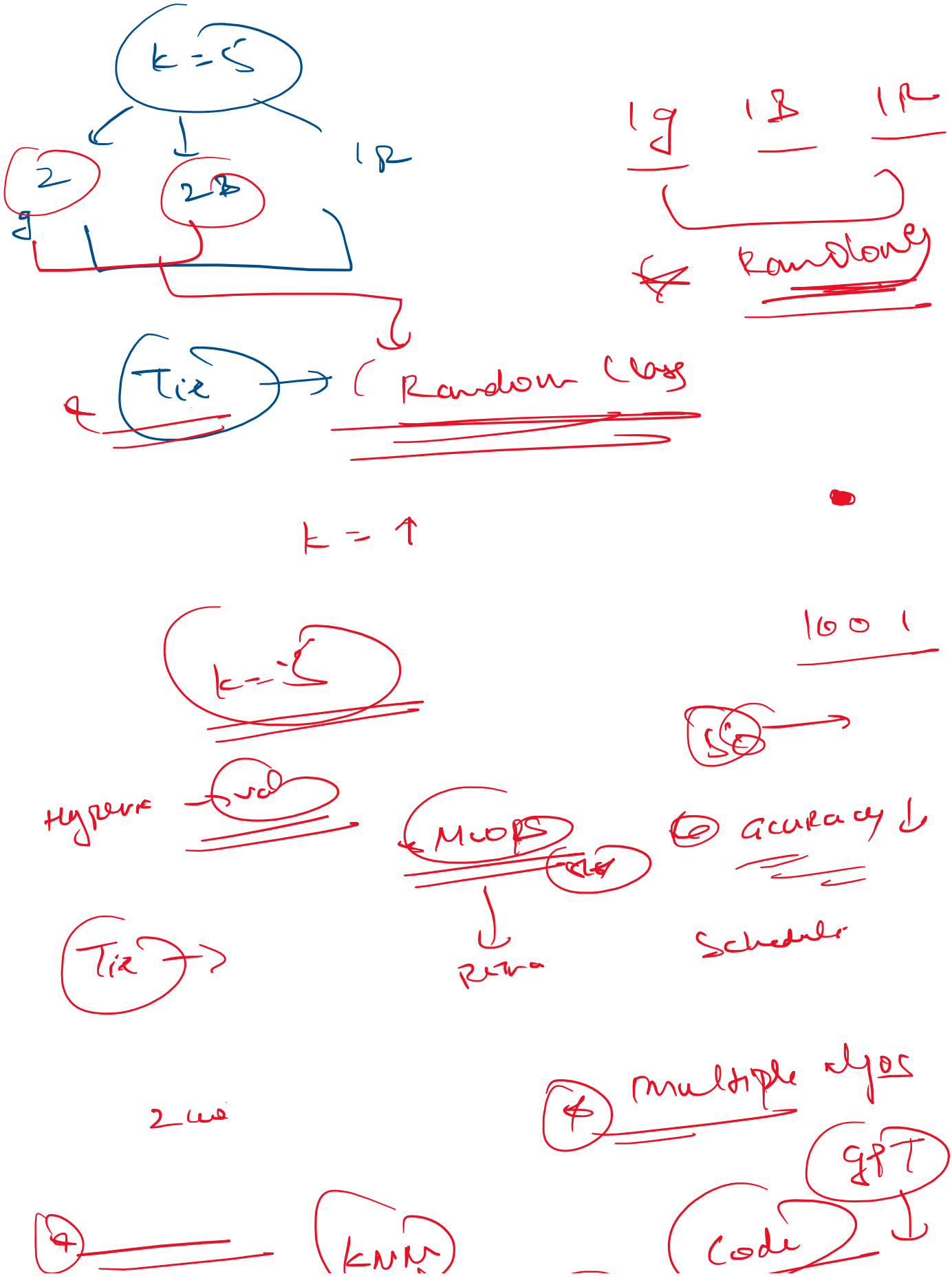
37 users have participated

A	s2,s1,s3,s4	0%
B	s2,s3,s4,s3	16%
✓ C	s2,s3,s1,s4	68%
D	s2,s4,s3,s1	16%

End Quiz Now





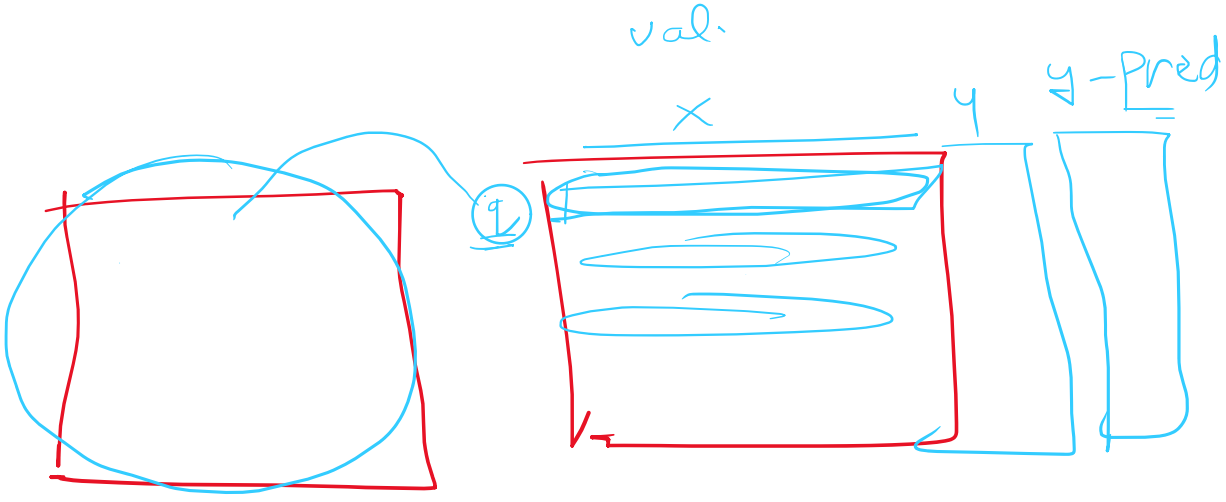

k = 5
2
2b
1R
1g
1B
1R
Random
Tie
Random Class
k = ↑
k = 5
100 1
hyperparameter
MLOPS
accuracy ↓
Tie
Scheduler
2 mo
multiple algos
KNN
gpt
code



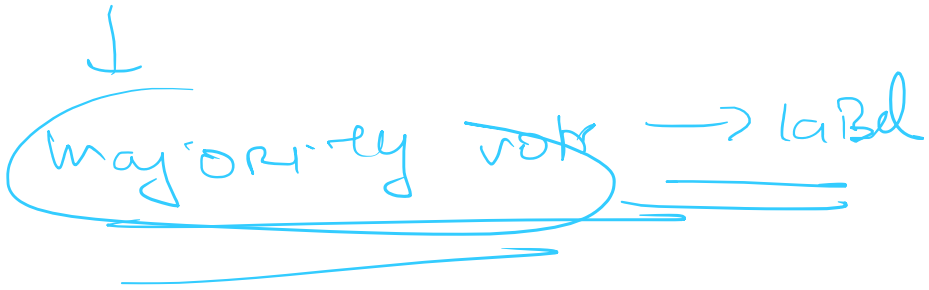
New DP

x_1 x_2

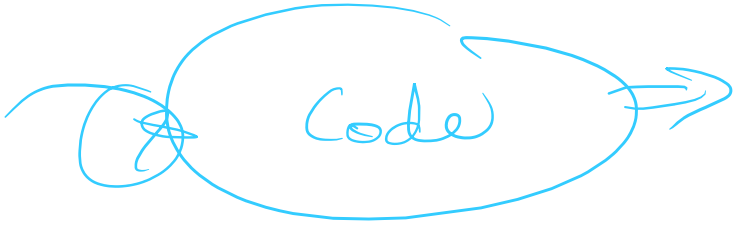
[4, 2] → dist. from all data











KNN → No assumption

lin Reg → Assumption

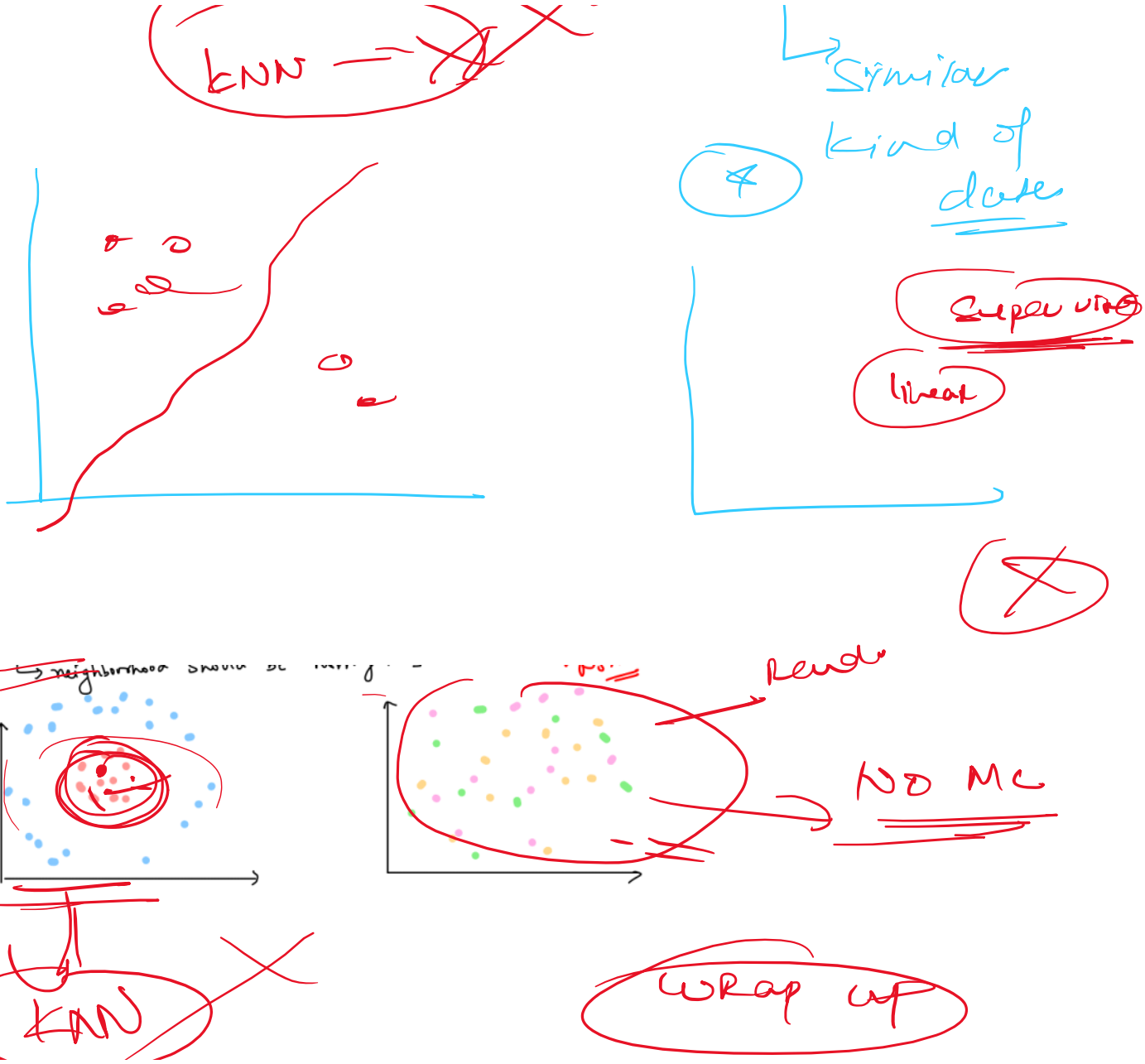
KNN → neighborhood

→ homogenous

di

KNN → condn

→ similar



Quiz time!

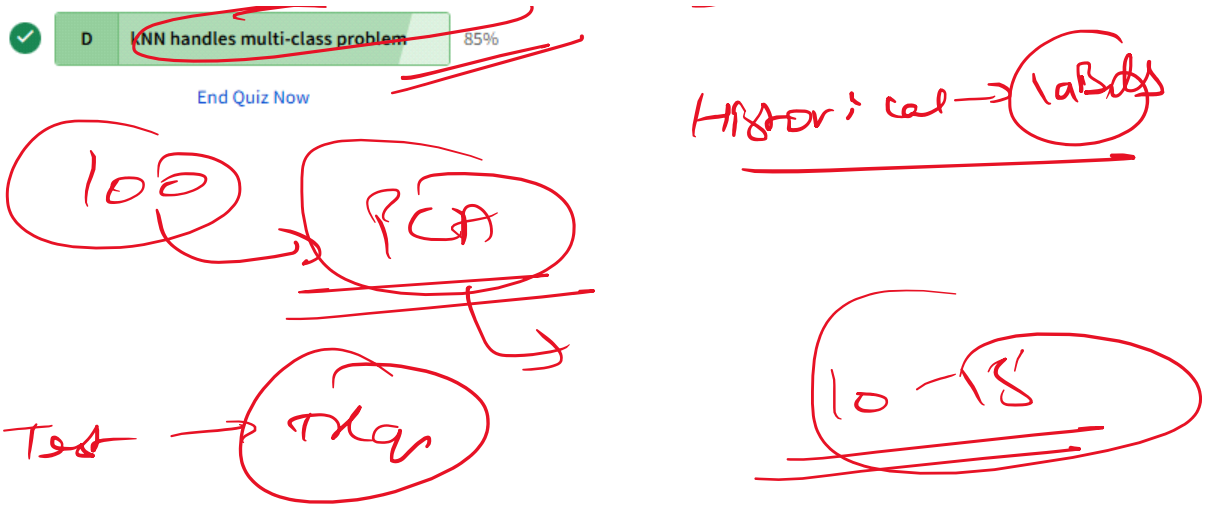
Time Left: 35s

how kNN is better than logistic regression. Select the correct option

26 users have participated

A	kNN has less time complexity	4%
B	kNN classifies data better	4%
C	kNN handles most noise/outlier	8%
D ✓	kNN handles multi-class problem	85%

End Quiz Now


D
KNN handles multi-class problem
85%
End Quiz Now